Building a CI/CD Pipeline for APIs



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Importance of CI/CD in API Management

- APIs are the de facto standard for connecting services, data and apps
- APIs are the pumping fuel of digitally driven organizations
- Rapid phase of API development
- Smooth propagation of APIs between environment is critical
- Ensure the functionality upon update across environments is important for the quality aspects

Why CI/CD in API Management?

- Rapid development and deployment of APIs
- Less human interruptions
- Fast delivery to end users
- Time saving and efficiency gain
- Automated process provide greater management flexibility
- Detect issues earlier

Building CI/CD Strategy

- Organizations follows variant strategies
- First step of building a CI/CD process associates with

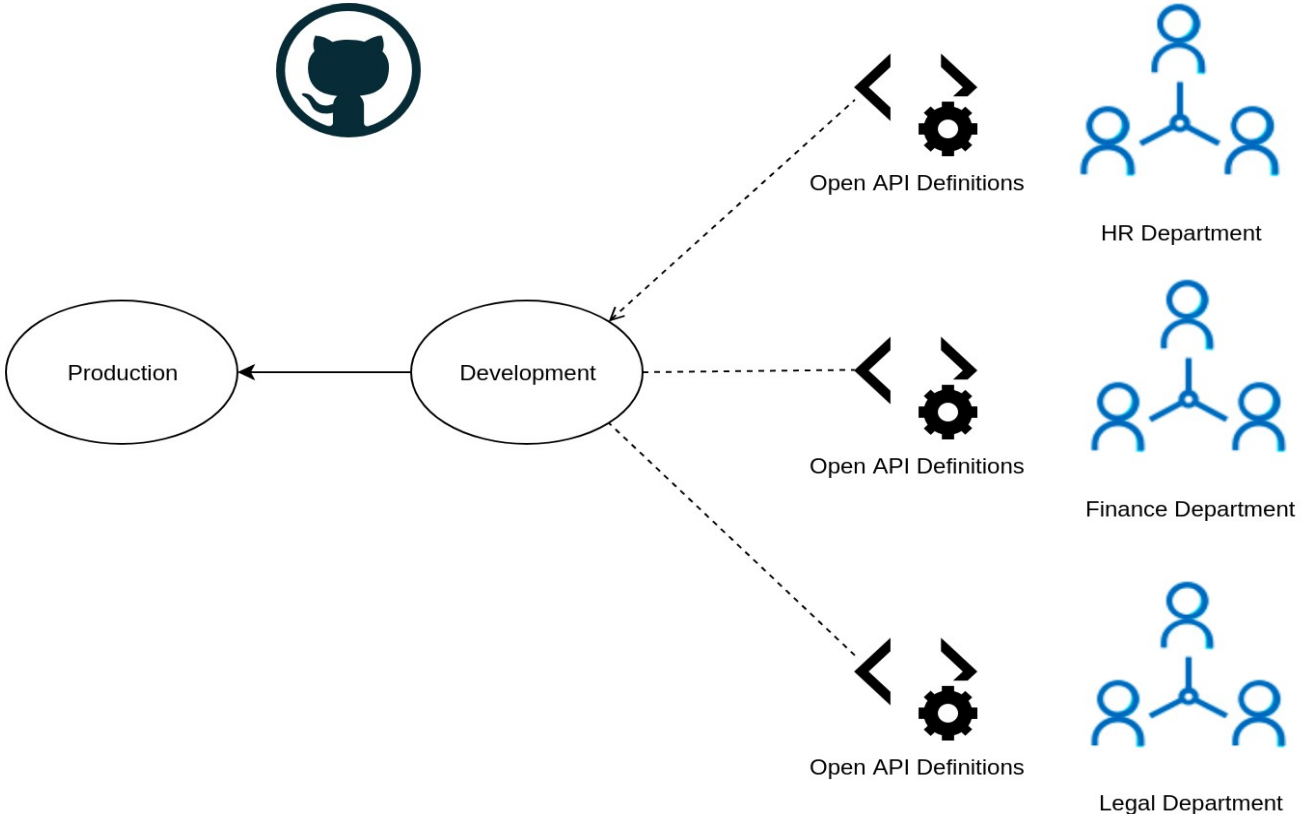
  Organization culture

  Governance structure and process

  Team structure and dynamics



Automation Challenges

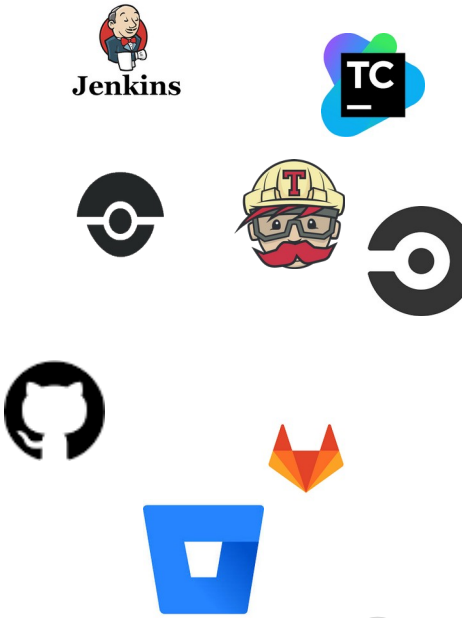


Automation Challenges

- Organizations are maintaining multiple deployment environments
- APIs associate with multiple policies and configurations
- Environment specific (endpoints) configurations
- Interference with multiple development teams

Automation Challenges

- Technology and tooling plays a vital role in the CI/CD process implementation
- Selection of process automation tool
    - Jenkins
    - Travis CI
    - TeamCity
    - Circleci
    - Drone
- Selection of source code repository
    - Github
    - Gitlab
    - Bitbucket



Support from API Management Vendor

- CI/CD approach of each API Management vendor is different
- API Vendors offer
    - Tooling support
    - Support for Interaction with source code repositories
    - Extensions to configure environment specific configurations
    - Plugins and extensions for pipeline tools
- Standard REST APIs

Configs that Differ Between Environments

- APIs contain environment specific configurations
- Some of the configurations differ between environments
- Environment-specific configurations include
    - Backend endpoints
    - Credentials of backend services
    - Certificates of endpoints
    - Endpoint timeout settings
    - Gateway environments

CI/CD with WSO2 API Manager

- Selection of process orchestration tool such as Jenkins is a key requirement
- CI/CD process requires a Source Code Management(SCM) system which acts as a Single Source of Truth for the pipeline
- apictl tool of API Manager provides the flexibility to integrate with these tools to migrate APIs between environments
- It is capable of handling environment specific configurations and can promote the API seamlessly to other environments via a single command

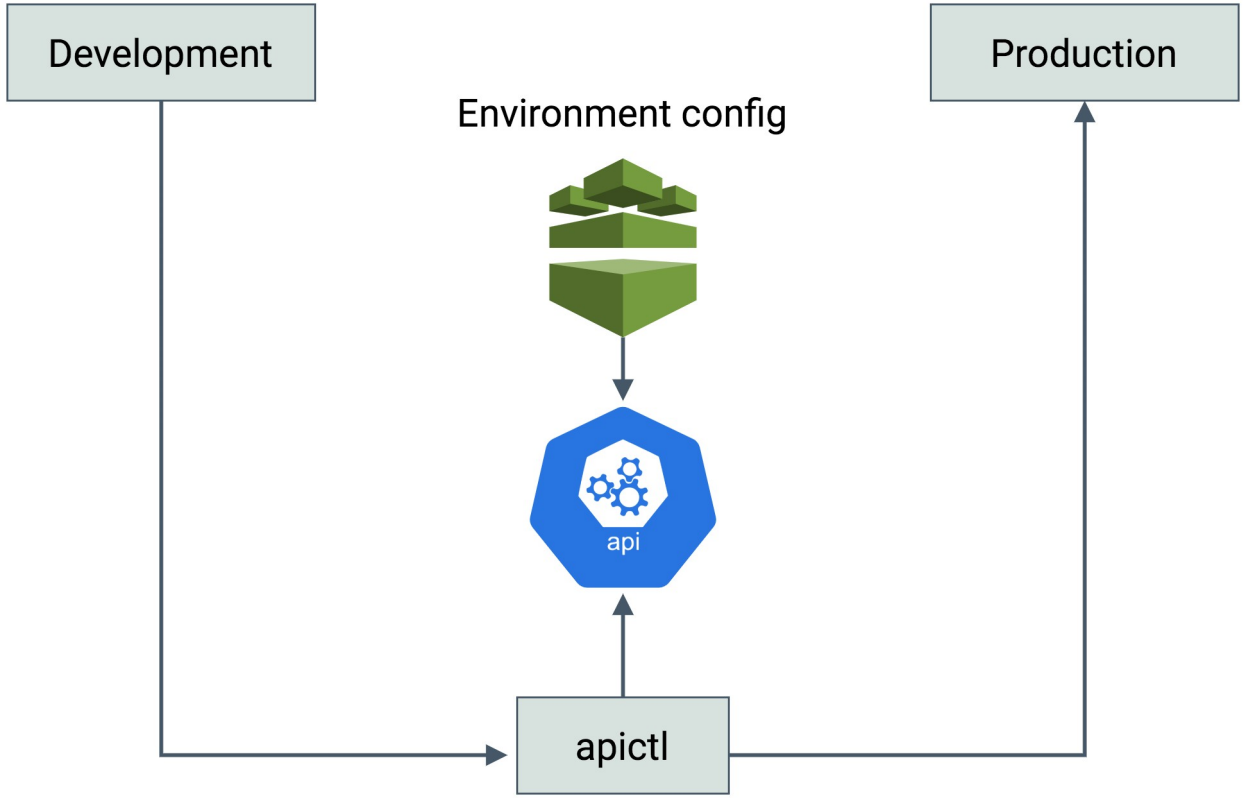
API Controller (apictl)

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APICTL and its Features

- Platform Agnostic
- Inject environment related configurations
- Command Line tool
- Easy configuration with many CI/CD tools

Simplistic View



CI/CD for APIs - Using the CLI

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Building Blocks for Creating a CI/CD Pipeline with CLI

- Prepare the environments
- Generate the API project
- Commit API project and deployment resources to a VCS for CI/CD
- Import the API to an upper environment

Preparing the Environments

- Install APIM-4.7.0
- Install apictl-4.7.0 and configure

Reference

Generate the API Project

- Create an API via API Publisher in a lower environment and export via apictl

or

- Initialize the project using a Swagger/OpenAPI specification

References [1]. [2], [3]

Extending a CI/CD Pipeline for Additional Operations

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Extending a CI/CD Pipeline for Additional Operations

- Using apictl in order to invoke an API or an API Product by subscribing to it using a new application created by apictl. [Reference]
- Extending a CI/CD pipeline to support API Products [Reference]
  - CI/CD pipeline can support API products other than APIs
  - The commands used are somewhat different and also a project for API product cannot be created using an OpenAPI definition.
- Extending a CI/CD pipeline to support applications [Reference]
  - Can use this CI/CD approach to export applications from lower environments to upper environments.

Let’s try it out!

Validating the WSO2 API Manager Environment

Building a CI/CD pipeline with GIT

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